

AI Lab Report Interpreter

Intelligent Analysis of Medical Diagnostics

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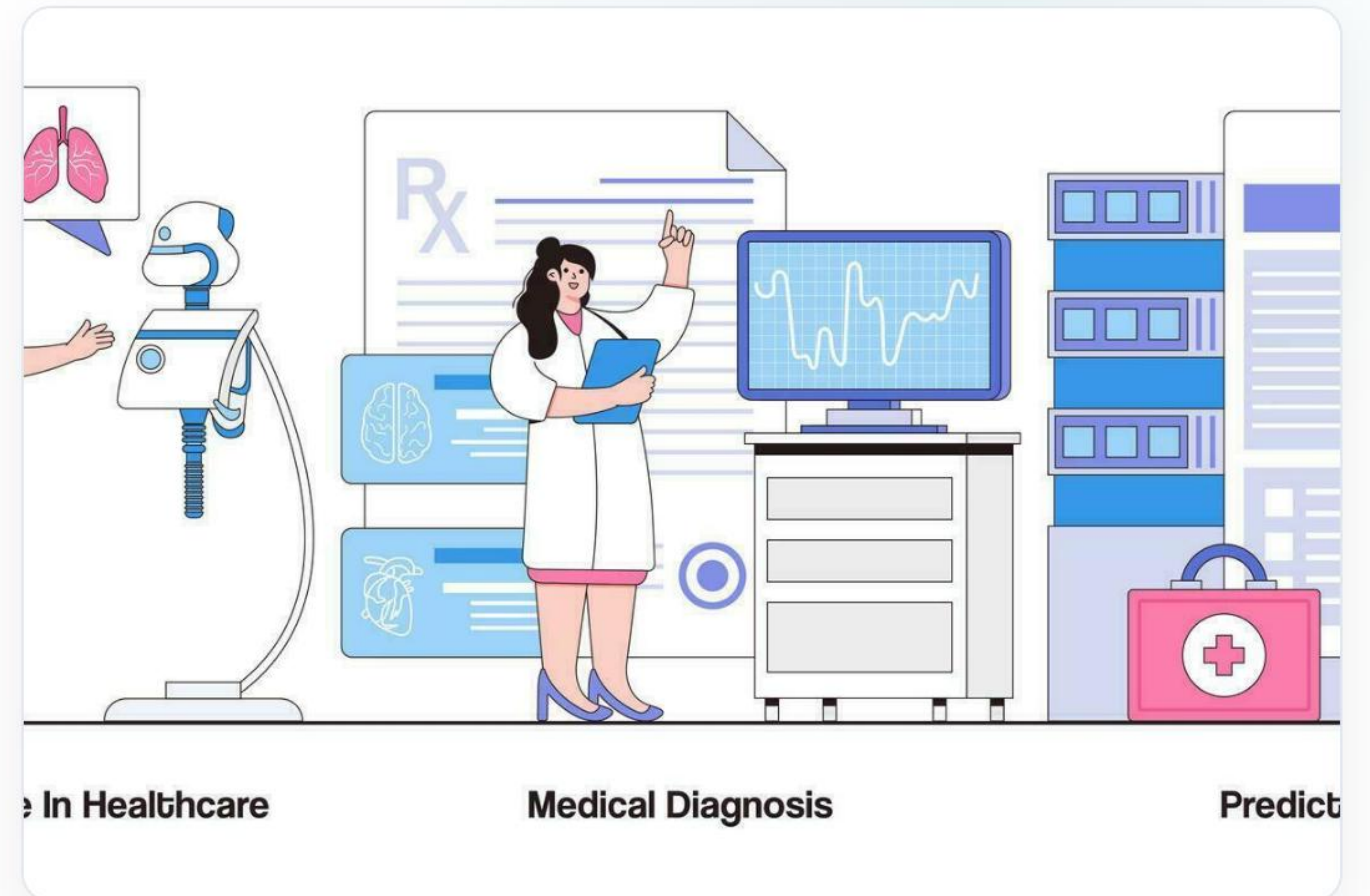
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| Project Overview

- ✓ **Simplifies Lab Data:** Automatically converts multi-format complex clinical lab reports into user-friendly insights.
- ✓ **Multi-Format Support:** Seamlessly ingests both image uploads and complex multi-page PDF documents.
- ✓ **Smart Extraction:** Uses industry-grade optical character recognition (OCR) to parse clinical terms.
- ✓ **Efficiency Booster:** Empowers healthcare professionals by heavily saving critical diagnostics verification time.



| The Industry Gap

The Problem: Manual review of medical lab data is highly slow, exhausting, and prone to severe operational human error. Our intelligent system removes slow manual entry pipelines completely, and significantly reduces the risks associated with human transcription misinterpretation. This ensures consistent diagnostic accuracy, liberating medical practitioners to focus fully on strategic face-to-face patient clinical interactions.



| Core Philosophy



Simplify

Condense complex diagnostic terminologies and biological readings into easy-to-digest, readable charts for stakeholders.



Automate

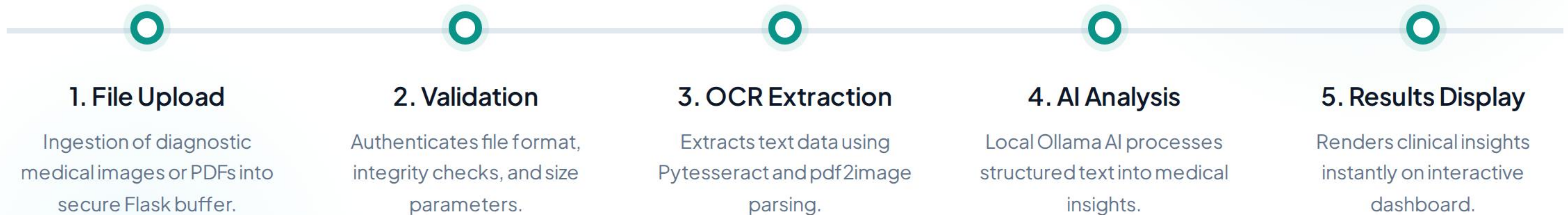
Eradicate physical administration delays. Automatically process, digitize, and file patient reports using smart local pipelines.



Accuracy

Deploy secure medical verification architectures to maintain precision and minimize clinical false positives.

Technical Workflow



| Tools & Technology



Flask

A highly secure and lightweight Python web framework optimized for swift routing and API deployment.



Tesseract

An advanced open-source OCR engine configured to read dense laboratory text tables perfectly.



Ollama

Underpins clinical intelligence locally, executing highly private and secure offline LLM model lookups.

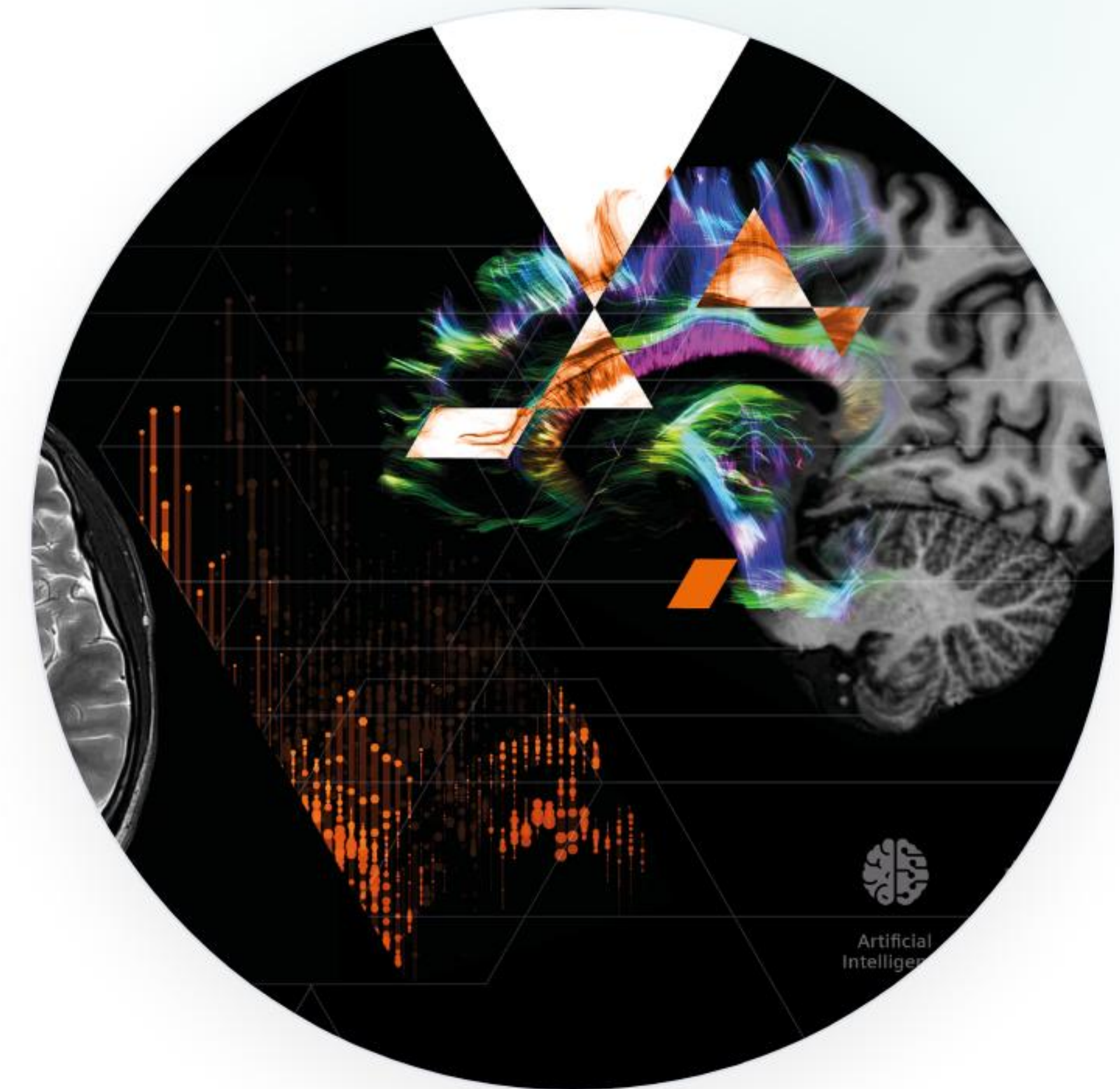


Medical KB

A specialized, custom built knowledge logic structure referencing validated diagnostic guidelines.

| AI Patient Assistant

- ✓ **Specialized Chatbot:** Accessible and interactive tool engineered specifically for direct clinical general medical dialogue.
- ✓ **Context-Aware Parsing:** Driven by local Ollama offline Large Language Models for robust natural phrasing.
- ✓ **Evidence-Based Design:** Strictly grounded using our pre-mapped structured Medical Knowledge base.
- ✓ **Friendly UX:** Features an inclusive, empathetic tone optimized for relieving patient health anxieties.



Data Management

Constraint	Specification Limit	Mechanism Used
Max File Size	16 Megabytes (MB)	Flask validation layer
Item History	50 Entry Max Cache	JSON rolling array logic
Data Pipeline	Extracted metadata output	Secure history.json
Timestamps	Automatic temporal hashes	System epoch logging

Secure, Private Storage

The architecture is completely optimized for private medical setups. By employing a rolling JSON cache file system offline, no third-party cloud is triggered during clinical parsing.

All temporary documents are auto-purged immediately following metadata processing, ensuring rigorous patient data confidentiality compliance.

Thank You!

Intelligent Analysis of Medical Diagnostics

Presented By Student Registrations:

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